

9.2 Lightweight components

The 1.1 API introduced the notion of lightweight components. Lightweight components directly extend either:

- `java.awt.Component`
- `java.awt.Container`

The `Component` class and its supporting cast are the foundation upon which the AWT is built. `Component` is an abstract base class for components such as menus, buttons, labels and lists. And `Container` is an abstract base class that extends `Component`. Classes derived from `Container` are

- `Panel`,
- `Applet`,
- `Window`,
- `Dialog` and
- `Frame`.

All can contain multiple components.

9.2.1 Component

It is an abstract class that provides a great deal of functionality for the classes that extend it. For example, a component has the following affiliated with it.

- Graphics object
- Location
- Size
- Native peer
- Parent container
- Fonts and font dimensions (font metrics)
- Foreground and Background colors
- Locale
- Minimum, Maximum and Preferred sizes

A component does not decide for itself how it will be shaped, and so its shape ends up taking a lot of abuse. Now, before you go blaming `Container` for this abuse, you must know that `Container` is an innocent bystander. Although a `Container` contains a `Container` , it is the `LayoutManager` who is 100% responsible for the abuse dished out to a `Container` . The `LayoutManager` decides the sizing and positioning of components. Using `Containers`, you can group related components and treat them as a unit. This simplifies an applet's design and is useful for arranging components on the display.